Ushasri Mogadali

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Summary

Software Engineer with 4 years of experience in full-stack development, IoT platforms, and DevOps practices. Skilled in Python, React.js, PostgreSQL, Docker with experience building production-grade applications, and real-time monitoring solutions. Co-authored multiple research publications and committed to continuous learning, professional development, and staying updated with latest technologies to drive organizational success.

Education

B.Tech in Computer Science *2018 - 2022*
Kakinada Institute of Engineering and Technology for Women, JNTUK University
CGPA: 7.85

Skills

Frontend: React.js, AEM, HTML, CSS, JavaScript
Backend: Python, FastAPI, Node.js
Database: PostgreSQL
DevOps: Git, Docker, Jenkins
Visualization: Grafana, Tableau
Tools: Postman, Locust, Figma, Canva, Blender

Experience

Senior Research Translation Engineer *Oct 2025 - Aug 2026*
IIIT Hyderabad, Smart City Research Center

- Contributed to the development and maintenance of a City IoT Operating Platform (ctOP) based on oneM2M standards, collaborating with academic, government, and enterprise stakeholders (IISc Bangalore, IUDX, MeitY).
- Implemented Git-based development workflows and Jenkins CI/CD pipelines to automate build, testing, and deployment processes.
- Led and mentor a team of 7 members, ensuring timely delivery of multiple projects through task management, code reviews, standardized documentation, workflow automation, and intern training.
- Co-authored two peer-reviewed research publications, demonstrating strong analytical skills and continuous learning.

Research Translation Engineer *Aug 2022 - Sep 2025*
IIIT Hyderabad, Smart City Research Center

- Developed and deployed a full-stack IoT dashboard for Wi-SUN–based streetlight control, supporting real-time monitoring of 60+ devices and geo-spatial analytics which is actively used by Silicon Labs.
- Designed and developed a transformer health monitoring dashboard using Tableau, with drill down capabilities, user management, and alerts and notifications, in collaboration with BYPL (Delhi) and GIZ.
- Conduct load testing and performance analysis using Locust to evaluate server utilization and optimize system throughput.
- Developed and deployed multiple IoT analytics dashboards providing real-time monitoring, visualization, drill down capabilities and operational insights for 300+ IoT devices across 10+ utility and environmental monitoring domains.
- Co-authored four peer-reviewed research publications, demonstrating strong analytical skills and continuous learning.

Projects

Net Zero Portal

- Designed and developed a React-based sustainability portal to track energy, water consumption, and waste generation using real-time IoT data.
- Developed and integrated AEM-based content management to enable users to create, update, and manage sustainability-related content, including research information, carbon accounting concepts, resource classifications.
- Developed and Integrated backend services and APIs to retrieve and present dynamic sustainability and IoT data.
- Built analytical dashboards to visualize sustainability metrics, resource consumption, emissions and reduction data.
- Researched carbon accounting standards, green vs. grey resource classification, and Scope 1, Scope 2, and Scope 3 emissions methodologies.
- Tech stack: React.js, AEM, Python, PostgreSQL, Grafana

Smart City IoT Dashboard

- Designed and developed an IoT dashboard for Wi-SUN–based smart streetlight systems, enabling real-time monitoring and control of 60+ connected devices.
- Built services for real-time device data ingestion, processing, and visualization.
- Implemented map based monitoring, analytics and operational insights for streetlight deployment.
- Developed live network topology visualization for users to monitor device connectivity in real time.
- Tech stack: React Js, PostgreSQL, Python, Grafana

Transformer Health Monitoring System

- Developed a proof-of-concept transformer health monitoring system using Tableau, based on energy parameters to assess transformer performance and health.
- Implemented time-interval filtering, user management, hierarchical drill-down analytics and threshold based monitoring features.
- Developed alert and notification system for anomaly and detection.
- Tech stack: React Js, Tableau, PostgreSQL, Python

CI/CD Pipeline and Monitoring System

- Designed and implemented an end-to-end CI/CD pipeline using Jenkins, automating build, testing, and deployment workflows.
- Containerized application services using Docker, enabling consistent and portable deployments across development, testing, and production environments.
- Integrated Grafana monitoring dashboards to track system performance, application health, resource utilization, and uptime.
- Tech stack: Git, Jenkins, Docker, Grafana

Publications

- **Insights into Smart City Solutions**, TSDSI Whitepaper. Link
- **WF-IoT 2023**, IEEE World Forum on Internet of Things. DOI: 10.1109/WF-IoT58464.2023.10539501
- **APSCON 2024**, IEEE Applied Sensing Conference. DOI: 10.1109/APSCON300364.2024.10465845
- **ICSA 2024**, Exploratory Study on oneM2M-based Interoperability. Link

Achievements

- **Awarded by Silicon Labs** for excellence in real-time data visualization and dashboard development.
- Represented the organization at **India Mobile Congress (IMC), New Delhi**, workshops, and multiple R&D exhibitions, presenting smart city, and sustainability solutions to industry, government, and academic stakeholders.
- Awarded **Best User Application** at the International **oneM2M Hackathon 2023** (organized by KETI,

South Korea) for developing a Smart Spaces solution using oneM2M.

Certifications

- Completed the **Wipro Velocity Program**, focused on enterprise software development and professional engineering practices.
- Awarded with **two-year Certification of Leadership Excellence** from KIET for leadership and team contribution.